VICTORIA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 1

CANDIDATE NAME ……………………………………………………………..

CT GROUP ……………………………..

CHEMISTRY **8873/02**

Paper 2 **11 September 2018**

Candidates answer on the Question Paper. **2 hours**

Additional Materials: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Answer **one** question.

The number of marks is given in brackets [ ] at the end of each question or part question.

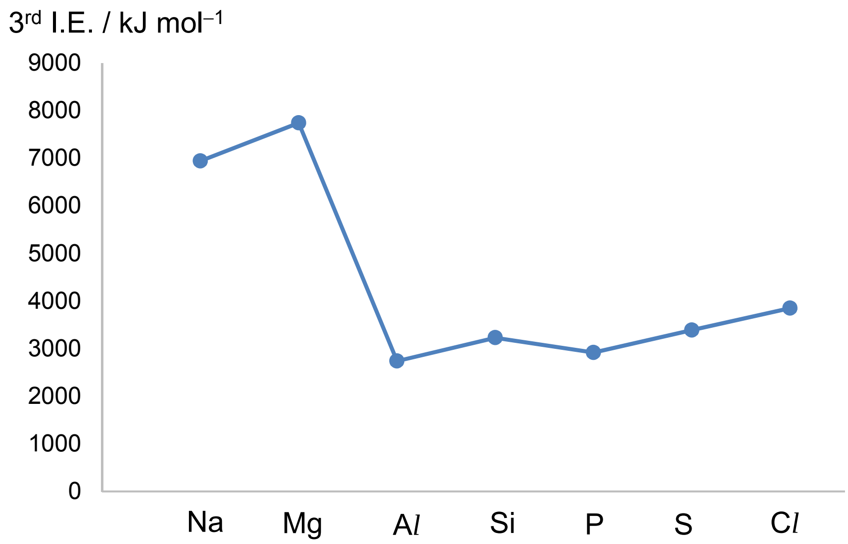
For Examiner's Use		
Section A	1	/ 20
	2	/ 12
	3	/ 8
	4	/ 20
Section B	5 / 6	/ 20
Total		/ 80

This document consists of **23** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section, in the spaces provided.

1 (a) The graph below shows the third ionisation energies of the Period 3 elements from Na to C*l*.



(i) With the aid of an equation, define the 3rd ionisation energy of phosphorous.

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.... [2]

(ii) Explain why there is a general increase in the 3rd ionisation energies from phosphorous to chlorine.

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.... [2]

(iii) Give the full electronic configuration of P^{2+}.

.... [1]

(iv) With reference to your answer from **(a)(iii)**, explain why the 3rd ionisation energy of Si is higher than that of P.

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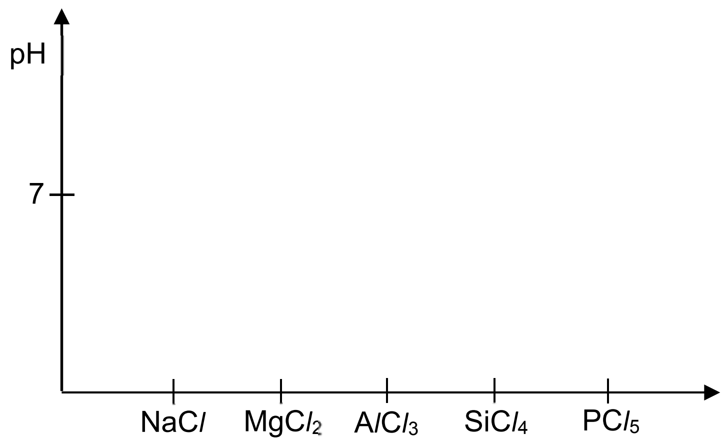
.... [2]

(b) **(i)** Sketch a graph on the axes provided to show the pH of the solutions produced when the Period 3 chlorides (from Na to P) are added to excess water.



[2]

(ii) Nitrogen is in the same group as phosphorous in the Periodic Table. Suggest whether NCl_5 is able to exist.

....

.... [1]

(iii) Arsenic is in the same group as phosphorous in the Periodic Table. With the aid of an equation, suggest the pH of the solution obtained when arsenic trichloride is added to excess water.

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.... [2]

(c) The boiling points of the two chlorides of phosphorous are shown in the table below.

compound	boiling point / ^{o}C
PCl_3	76.1
PCl_5	166.8

(i) Explain the difference in boiling points between the two chlorides of phosphorous.

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(ii) PCl_3 is able to react with BF_3. State the type of bond formed and draw a diagram to illustrate the shape of the product obtained.

[2]

(d) In the solid state, PCl_5 exists as an ionic lattice with the formula $[PCl_4]^+[PCl_6]^-$.

(i) State the shapes and bond angles of the $[PCl_4]^+$ and $[PCl_6]^-$ ions.

ion	shape	bond angle
$[PCl_4]^+$		
$[PCl_6]^-$		

[2]

(ii) Compare the lattice energy of $[PCl_4]^+[PCl_6]^-$ with that of NaC*l*. Explain your reasoning.

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[Total: 20]

2 **(a)** 3–chloropropanoic acid, $CH_2ClCH_2CO_2H$, is a weak Bronsted acid. Solution **A** is a 0.100 mol dm^{-3} $CH_2ClCH_2CO_2H$ solution. It has a pH of 2.49.

(i) Write the expression for the K_a of 3–chloropropanoic acid, including the units.

[1]

(ii) Hence, calculate the pK_a of 3–chloropropanoic acid.

[2]

(iii) When 25 cm^3 of 0.0500 mol dm^{-3} KOH was added to 25.0 cm^3 of solution **A**, a buffer solution is formed. With the aid of two equations, explain how the buffer solution is able to maintain a relatively constant pH.

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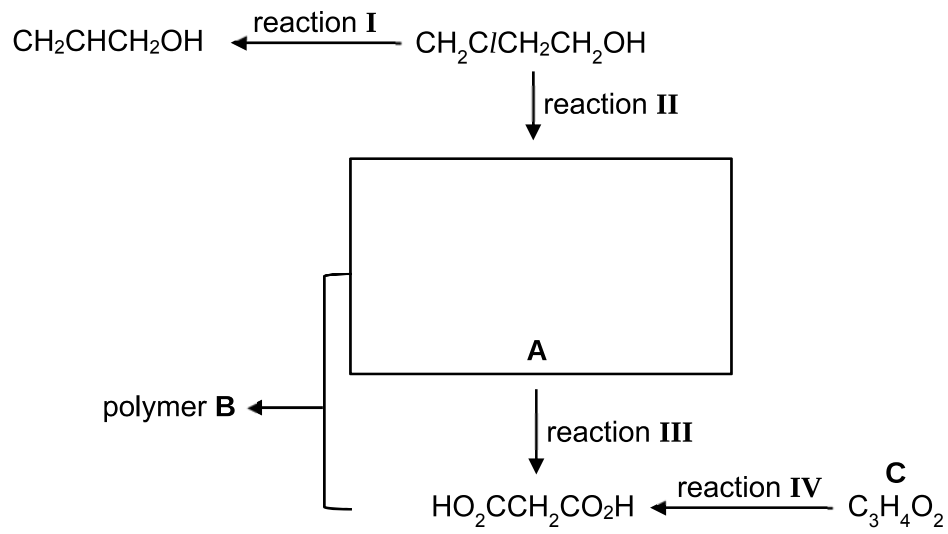
(b) 3–chloropropanoic acid can undergo a series of reactions as shown below.



(i) Draw the full structural formula of compound **A** in the box. [1]

(ii) State the reagents and conditions required for reactions **I**, **II** and **III**.

Reaction **I**:

reagents ....

condition ....

Reaction **II**:

reagents ....

condition ....

Reaction **III**:

reagents ....

condition ....

[3]

(iii) Draw the structure of one repeat unit of the polymer **B**.

[1]

(iv) Suggest the structural formula of compound **C**.

[1]

[Total: 12]

3 **(a)** Single–walled carbon nanotubes behave like cylindrical graphene. Multi–walled carbon nanotubes consist of multiple–rolled layers of graphene.

Single-walled carbon nanotube

Multi-walled carbon nanotube

(i) Plastic is a good replacement for steel because of its low cost. The insertion of carbon nanotubes into plastic has made it at least 10 times stronger than steel.

Describe the structure and bonding of a carbon nanotube. Hence, explain whether a single–walled carbon nanotube or a multi–walled carbon nanotube has higher tensile strength.

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(ii) Similar to graphene, carbon nanotubes conduct electricity. They can be ‘unzipped’ to form sheets of oxidised graphene as shown in **Figure 1**.

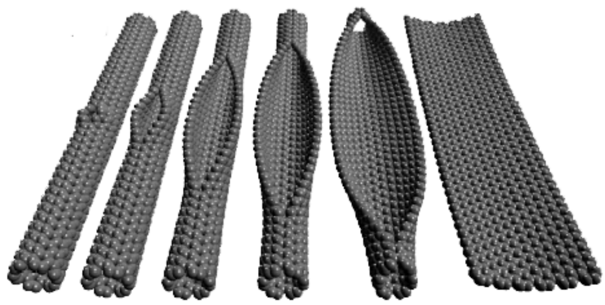

Figure 1

The ‘unzipping’ of a carbon nanotube to form oxidised graphene at the molecular level is shown in **Figure 2** below.

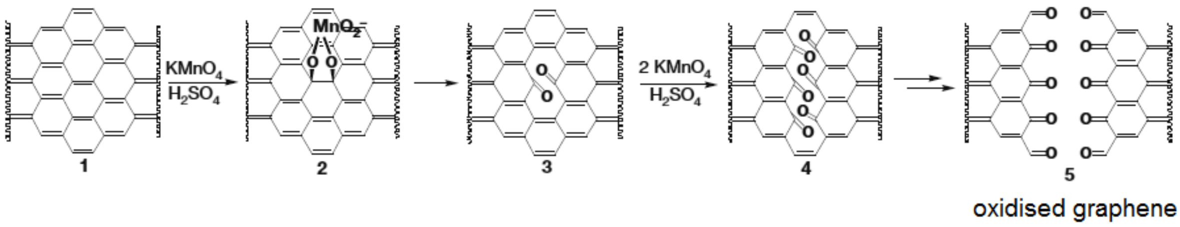

Figure 2

With reference to **Figures 1** and **2**, explain whether sheets of oxidised graphene can conduct electricity.

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(iii) We can obtain a sheet of graphene by reduction of oxidised graphene, only if the width is less than 50 nm.

Currently, it is not possible to produce very large sheets of graphene. Suggest how the edges of the graphene sheet pose difficulties to the production of large sheets of graphene.

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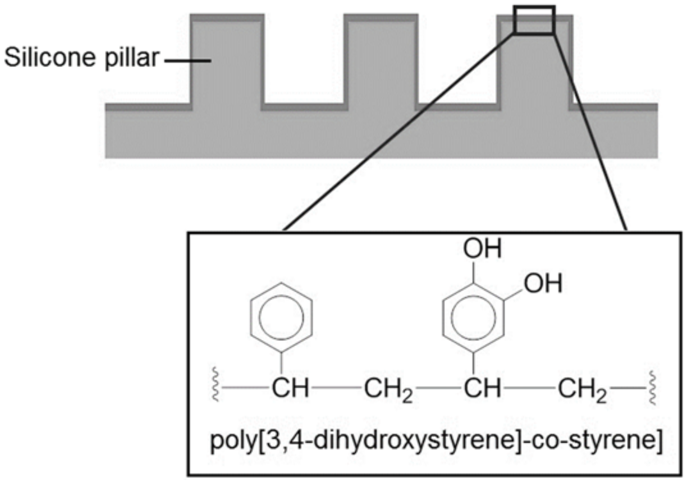
(b) Geckos are lizards with the remarkable ability to scamper up walls. A shortcoming of synthetic gecko–inspired fibres is that it cannot operate on both wet and dry surfaces.

Researchers have created 'geckel' which is an array of high silicone pillars sitting on a solid substrate. Each cylindrical pillar has dimensions of 400 nm in diameter and 600 nm in height. The surfaces of the pillars are coated with a synthetic adhesive polymer, poly[3,4–dihydroxystyrene]–co–styrene], as shown below. 'Geckel' can be used to create wall–climbing robots.



(i) Explain whether each silicone pillar can be considered as a nanomaterial.

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…………………………………………………………………………………………… [1]

(ii) A wall has been painted with a paint comprising of a mixture of hydrocarbons while another wall has been painted with a paint comprising of a polymer, poly(ethylene oxide).

$$H\!-\!\!\left[OCH_2CH_2\right]_n\!\!-\!OH$$

poly(ethylene oxide)

Explain how the silicone pillars and synthetic adhesive polymer of 'Geckel' enable a wall–climbing robot to support its own weight on both types of paint surfaces.

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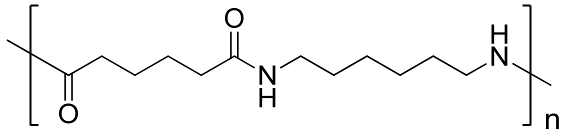
[Total: 8]

4 (a) The two most common types of nylons used in textile and plastic industries are nylon−6 and nylon−6,6. The structure of nylon−6,6 is shown below.



nylon-6,6

(i) Define the term *polymer*.

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(ii) Ropes made of nylon−6,6 are chosen for its high tensile strength. Ropes made from polyesters such as poly(ethylene terephthalate) (PET) are about 90% as strong as that of nylon−6,6.

With the aid of a diagram, showing relevant interactions, explain why nylon−6,6 is stronger than polyester.

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(iii) Rust or grout on toilet tiles can be removed by spraying the tiles with a rust remover (**Figure 3**) before scrubbing the tiles with a brush. Explain whether a brush with nylon bristles or high density poly(ethene) bristles should be used for the scrubbing.

Figure 3

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…………………………………………………………………………………………… [2]

(iv) State whether nylon−6,6 is classified as a thermoplastic or thermoset. Hence, explain whether it can be recycled.

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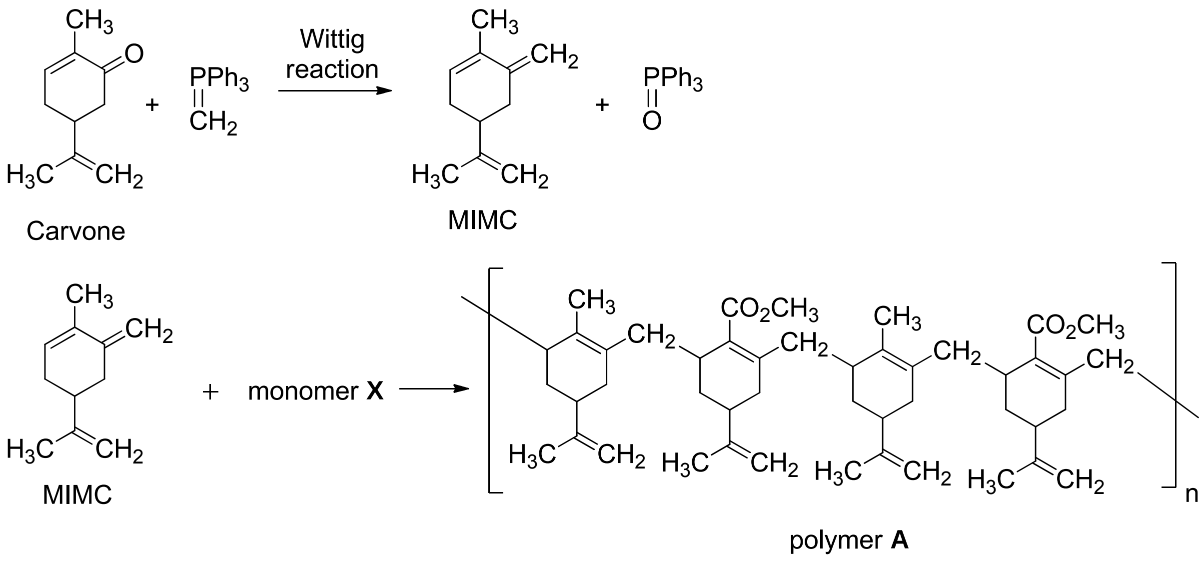
(b) Cavone is found in the oils of spearmint. It undergoes a Wittig reaction to yield the monomer, MIMC, which could undergo polymerisation with monomer **X** to form polymer **A**.



(i) State the type of polymerisation that has occurred to form polymer **A**. Hence, draw the constitutional formula of monomer **X**.

[2]

(ii) Explain why polymer **A** has a more rigid structure as compared to nylon−6,6 in **(a)**.

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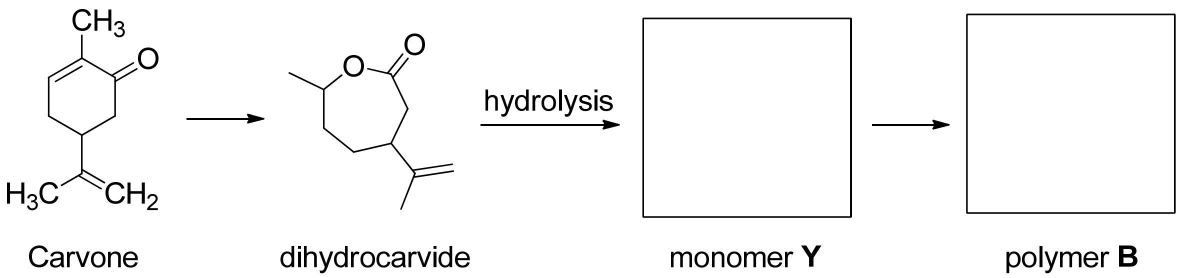
.... [1]

Carvone can also be transformed into dihydrocarvide which undergoes a series of steps to form polymer **B**.



(iii) Write an equation for the base hydrolysis of dihydrocarvide.

[1]

(iv) Infra–red spectroscopy can be used to identify functional groups present in organic compounds. For example, the ketone functional group of carvone shows an absorption signal at 1670–1740 cm^{-1}.

Polymer **B** shows the following infra-red absorption signals at 1635–1690 cm^{-1}, 1050–1330 cm^{-1} and 1730–1750 cm^{-1}.

With reference to the *Data Booklet*, identify the bonds that correspond to the above infrared absorptions.

1635–1690 cm^{-1} ……………………………………………………………

1050–1330 cm^{-1} ……………………………………………………………

1730–1750 cm^{-1} ……………………………………………………………

[1]

(v) With the aid of an equation, define the *enthalpy change of combustion* of carvone, $C_{10}H_{14}O$.

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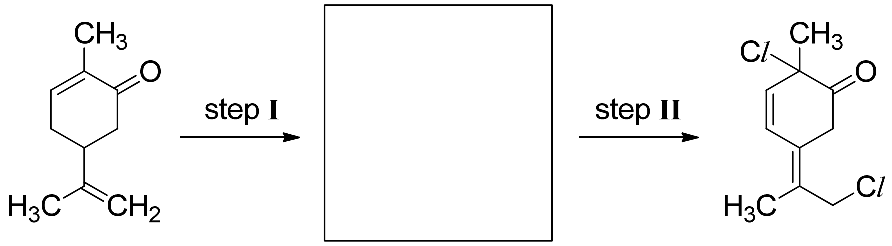
Equation:

[2]

(vi) Draw the structure of compound **Z** in the box for the following synthesis. State the reagents and conditions for step **I** and step **II**.



Carvone

compound **Z**

Step **I**:

reagents ………………………………………………….

condition ………………………………………………….

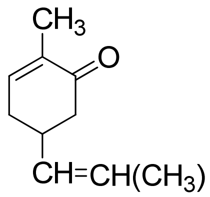
Step **II**:

reagents ………………………………………………….

condition ………………………………………………….

[3]

(vii) Only one of the C=C bonds in the isomer of carvone exhibits *cis–trans* isomerism. Draw the skeletal structures of the *cis–trans* isomers.



isomer of carvone

[2]

[Total: 20]

Section B

Answer **one** question from this section, in the spaces provided.

5 (a) Phosphine has the chemical formula PH_3. The kinetics of the reaction between phosphine and sodium chlorate(I) can be investigated by the following method.

$$PH_3 + 2NaClO \rightarrow H_3PO_2 + 2NaCl$$

Experiment	Initial concentrations of reactants / mol dm^{-3}		Initial rate of disappearance of PH_3 / mol dm^{-3} s^{-1}
	PH_3	NaC*l*O	
1	0.24	3.50	3.68
2	0.14	5.70	3.58
3	0.48	3.50	7.36

(i) Determine the order of reaction with respect to PH_3 and NaC*l*O.

[2]

(ii) Write the rate equation for this reaction.

.... [1]

(iii) Determine the rate constant for experiment 1 and state its units.

[2]

(iv) Calculate the initial rate of formation of NaC*l* when the initial concentrations of PH_3 and NaC*l*O are 0.36 mol dm^{-3} and 2.50 mol dm^{-3} respectively.

[1]

(v) Determine the concentration of NaC*l* produced after two half-lives for experiment 1.

[2]

(b) The Appel reaction is used to convert alcohols to alkyl halides while triphenylphosphine, PPh_3, is oxidised to triphenylphosphine oxide, $O{=}PPh_3$.

Ph in PPh_3 and $O{=}PPh_3$ is a phenyl group, which remains unchanged after the reaction.

$$PPh_3 + CBr_4 + CH_3CH_2OH \rightarrow O{=}PPh_3 + CHBr_3 + CH_3CH_2Br$$

(i) With reference to the *Data Booklet*, calculate the enthalpy change for this reaction.

[2]

(ii) Using bond energy values from the *Data Booklet*, explain why the rate of reaction is slower when CCl_4 is used in place of CBr_4.

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(iii) Triphenylphosphine. PPh_3 can also be oxidised by oxygen to form triphenylphosphine oxide, $O{=}PPh_3$ with the use of a catalyst $[MoO_2Cl_2(DMSO)_2]$.

With the aid of a Maxwell–Boltzmann distribution diagram, explain how the addition of a catalyst speeds up the rate of reaction.

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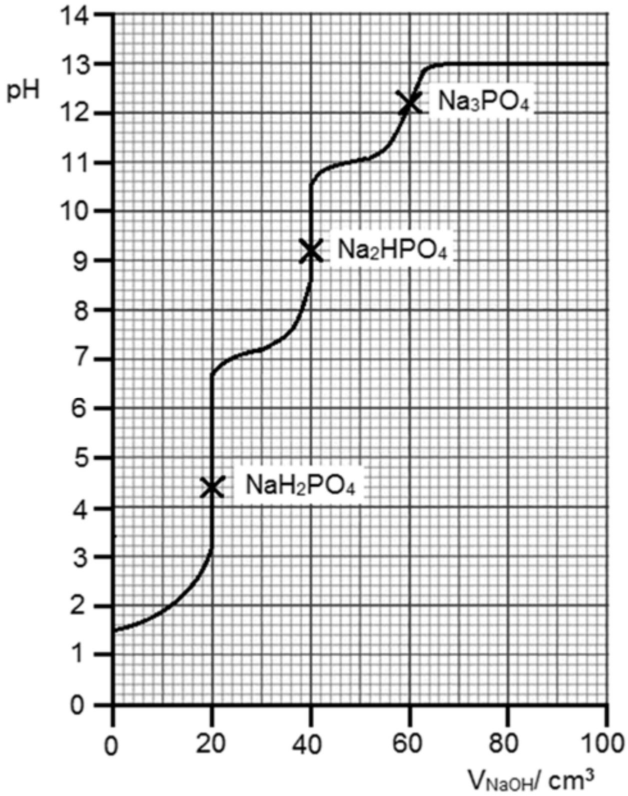
(c) In a titration experiment, 0.100 mol dm^{-3} NaOH was added to 15.0 cm^3 of phosphoric acid as shown below.



Phosphoric acid is a tribasic acid, and it dissociates into PO_4^{3-} in three steps as shown below.

Step 1: $H_3PO_4 \rightleftharpoons H_2PO_4^- + H^+$ $\quad K_{a1}$
Step 2: $H_2PO_4^- \rightleftharpoons HPO_4^{2-} + H^+$ $\quad K_{a2}$
Step 3: $HPO_4^{2-} \rightleftharpoons PO_4^{3-} + H^+$ $\quad K_{a3}$

(i) Calculate the concentration of the phosphoric acid.

[1]

(ii) With reference to the titration curve and **(c)(i)**, suggest whether phosphoric acid is a weak or strong acid.

[2]

(iii) Arrange the three K_a values of phosphoric acid in decreasing order of acidity. Explain your answer.

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.... [2]

(iv) The table below lists some common indicators found in the college laboratory. With reference to the given titration curve for phosphoric acid, suggest an indicator to detect the second end–point. Hence, state the colour change of your chosen indicator at the second end-point of the titration.

indicator	pH range for colour change	colour in acidic pH	colour in basic pH
methyl orange	3–5	red	yellow
bromothymol blue	6–8	yellow	blue
phenolphthalein	8–10	colourless	pink
thymolphthalein	9.3–10.5	colourless	blue
trinitrobenzoic acid	12–13.4	colourless	orange–red

....

.... [1]

[Total: 20]

6 (a) During an experiment, a student added 3.00 g of solid KOH to 100 cm^3 of 1.00 mol dm^{-3} HNO_3 solution. The initial temperature of the HNO_3 solution was 29.2 °C and the highest temperature recorded was 36.5 °C.

(i) Define the *standard enthalpy change of neutralisation,* $\Delta H^{\ominus}_{neut}$.

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(ii) Calculate the ΔH_{neut} of this reaction. You may assume that 4.18 J of energy is required to raise the temperature of 1 g of the solution by 1 °C.

[2]

(b) Compressed Natural Gas (CNG) has been used as a fuel for vehicles in Singapore since 2001 because it releases less air–borne pollutants. Together with the mandatory catalytic converters installed in vehicles, this helps to maintain the high air quality that Singapore enjoys.

The reaction below takes place in the catalytic converter and it is catalysed by a rhodium catalyst.

$$2CO(g) + 2NO(g) \rightarrow N_2(g) + 2CO_2(g) \qquad \Delta H < 0$$

(i) Draw energy profile diagrams for both the catalysed and uncatalysed reactions on the axes below. Label on the diagrams the axes, ΔH, activation energies for both the catalysed (E_a(cat)) and uncatalysed (E_a) reactions.

[2]

(ii) Explain in terms of collision theory, how the addition of a catalyst increases the rate of reaction.

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(iii) Explain the mode of action of the rhodium catalyst in this reaction.

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(c) Natural gas has many impurities such as CH_4 and H_2S, which can react according to the following equilibrium.

$$CH_4(g) + 2H_2S(g) \rightleftharpoons CS_2(g) + 4H_2(g)$$

2.0 mol of CH_4, 4.0 mol of H_2S, 2.0 mol of CS_2 and 6.8 mol of H_2 were allowed to reach equilibrium in a 250 cm^3 vessel at 800 ^oC. The equilibrium concentration of CH_4 was found to be 3.4 mol dm^{-3}.

(i) Calculate the value of K_c, include the units in your answer.

[2]

(ii) Define *Le Chatelier's Principle*.

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(iii) Explain how the position of equilibrium would be affected if the volume of the vessel is changed to 500 cm^3.

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(iv) Explain how the position of equilibrium would be affected if a catalyst was introduced into the vessel.

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(d) Hydrogen peroxide and iodide ions react in an acidic medium to form water and iodine as shown below.

$$H_2O_2 + 2I^- + 2H^+ \rightarrow 2H_2O + I_2$$

The rate of reaction is monitored by measuring the amount of iodine produced with time. The concentration of H_2O_2 remaining is then calculated as shown in the table below.

Time / s	$[H_2O_2]$ / mol dm^{-3}
0	0.0200
80	0.0167
180	0.0135
315	0.0103
490	0.0070
760	0.0038

(i) Plot a graph of [H_2O_2] against time on the axes below. Hence, determine the order of reaction with respect to [H_2O_2].

[3]

(ii) Use the graph to determine the initial rate of the reaction.

[2]

[Total: 20]

End of Paper